

Synthetic Native Template

Redistributable test fixture — not
laboratory data

Representative content layout

- Native title/content placeholders

Progress / Previous Commitments

Current position | H002

Commitment | NS101 | done | Owner: Gary | Due: 2026-09-10T09:00:00Z

Dependencies | pressure fixture | Parallel: True

Commitment | NS201 | planned | Owner: Gary | Due: 2026-09-24T00:00:00Z

Dependencies | pressure fixture | Parallel: True

H001 | Hypothesis

Hypothesis | Bulk conductivity drives the positional signal gradient.

Falsifier | Matched conditions do not change the predicted CV or resistance.

Research question | 提升 bulk conductivity 是否足以降低位置依賴的訊號變異？

H001 | Problem

Problem | 合成薄膜的缺陷密度沿位置增加，且訊號重複性下降。

Previous finding | 含水量與導電度存在位置梯度

Conflict | 導電度提高是否足以修復重複性仍未知。

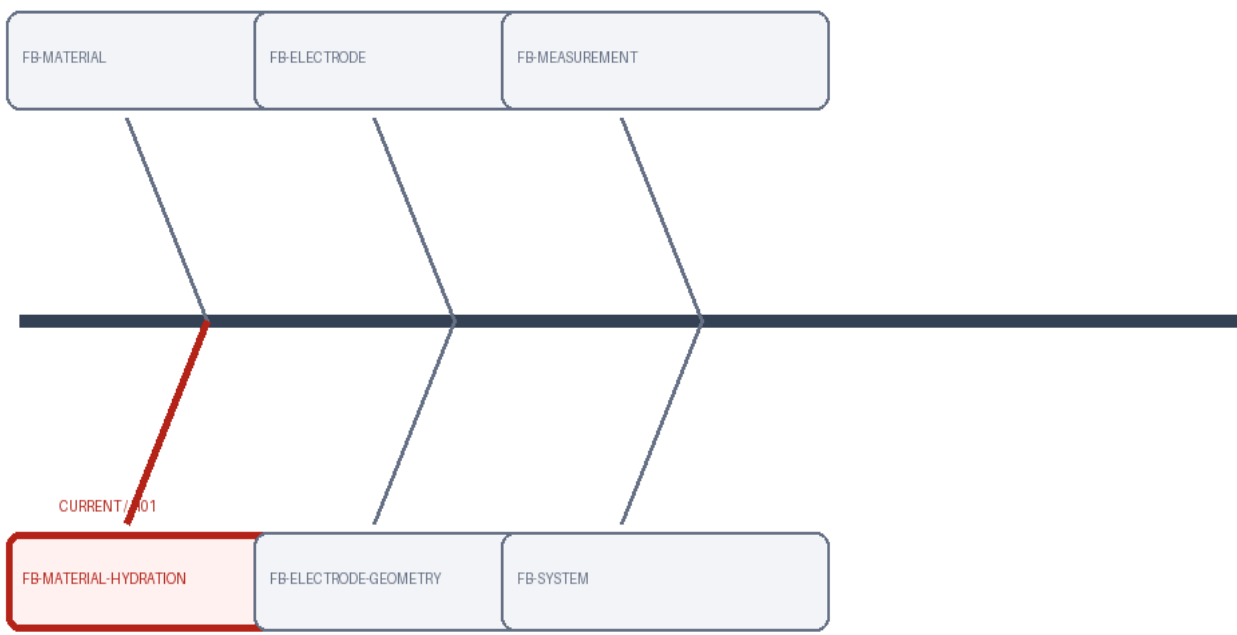
Research question | 提升 bulk conductivity 是否足以降低位置依賴的訊號變異？

Scope | 合成五個位置、每個位置三個 replicate 的機制辨識。

H001 | Total Fishbone / Research

Map

Historical snapshot |
FB001 rev1
Focus branch | FB-
MATERIAL-HYDRATION
Immutable map; current
branch highlighted.



H001 | Observation → Literature → Mechanism

Observation | 位置依賴缺陷與訊號變異已觀察到。

Problem | 現有模型無法解釋此變異。

Research question | 提升 bulk conductivity 是否足以降低位置依賴的訊號變異？

Literature consensus | transport gradient 可產生位置效應。

Alternatives | interface contact remains alternative

Gap | 缺少控制比較隔離機制。

Implication | 需匹配條件後比較。

Mechanism | bulk transport

Strategy | 均化導電度

SYNTHETIC OBSERVATION

H001 | Literature → Mechanism

→ Strategy

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Problem | 現有模型無法解釋此變異。

Research question | 提升 bulk conductivity 是否足以降低位置依賴的訊號變異？

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Mechanism | bulk transport

Strategy | 均化導電度

H001 | Experiment Design

EXP101 | IV: ['含水量']

Controls: ['原始配方']

N/replicates: {'replicates': 3, 'samples': 15}

Metrics/units: ['導電度 (mS/cm)']

Method: ['synthetic-four-probe']

Prediction: ['導電度提升']

Decision rule: {'go': 'mean conductivity increases $\geq 20\%$ ', 'partial_go': '10-20%', 'no_go': '<10%'}

EXP102 | IV: ['位置']

Controls: ['原始位置分布']

N/replicates: {'replicates': 3, 'samples': 15}

Metrics/units: ['訊號 CV (%)']

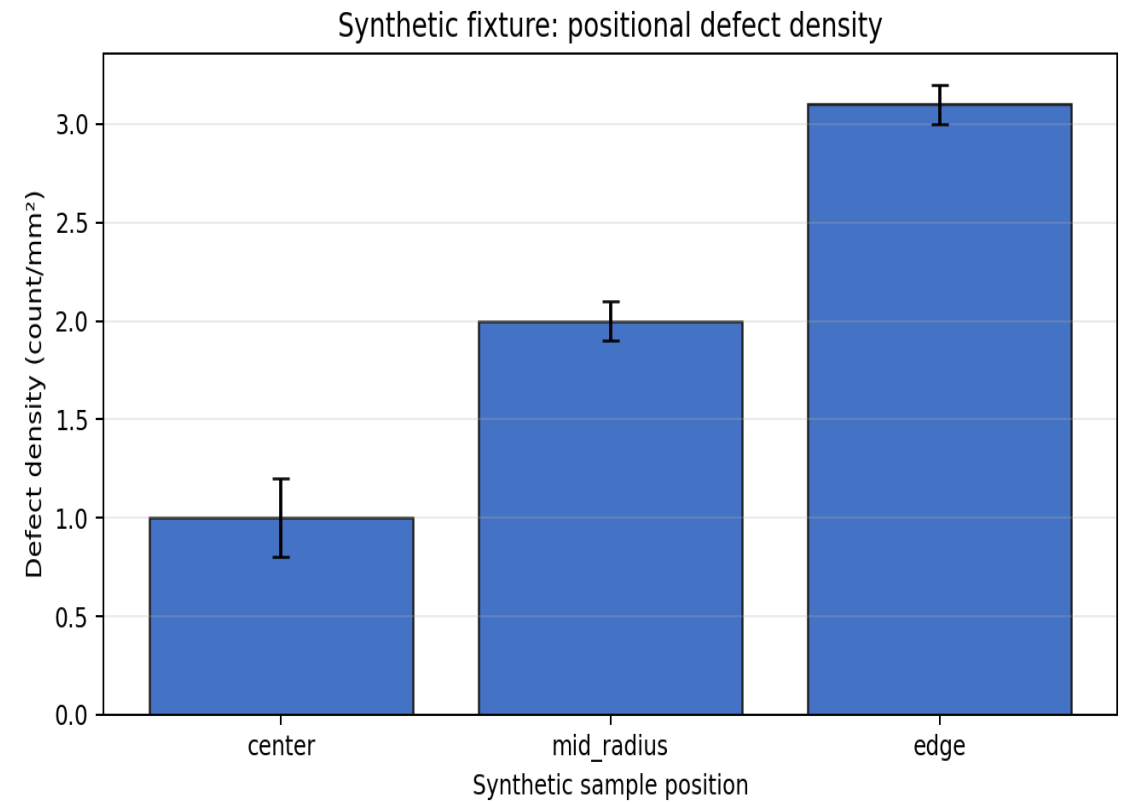
Method: ['synthetic-signal-test']

Prediction: ['CV 下降']

Decision rule: {'go': 'CV decreases $\geq 30\%$ ', 'partial_go': '10-30%', 'no_go': '<10%'}

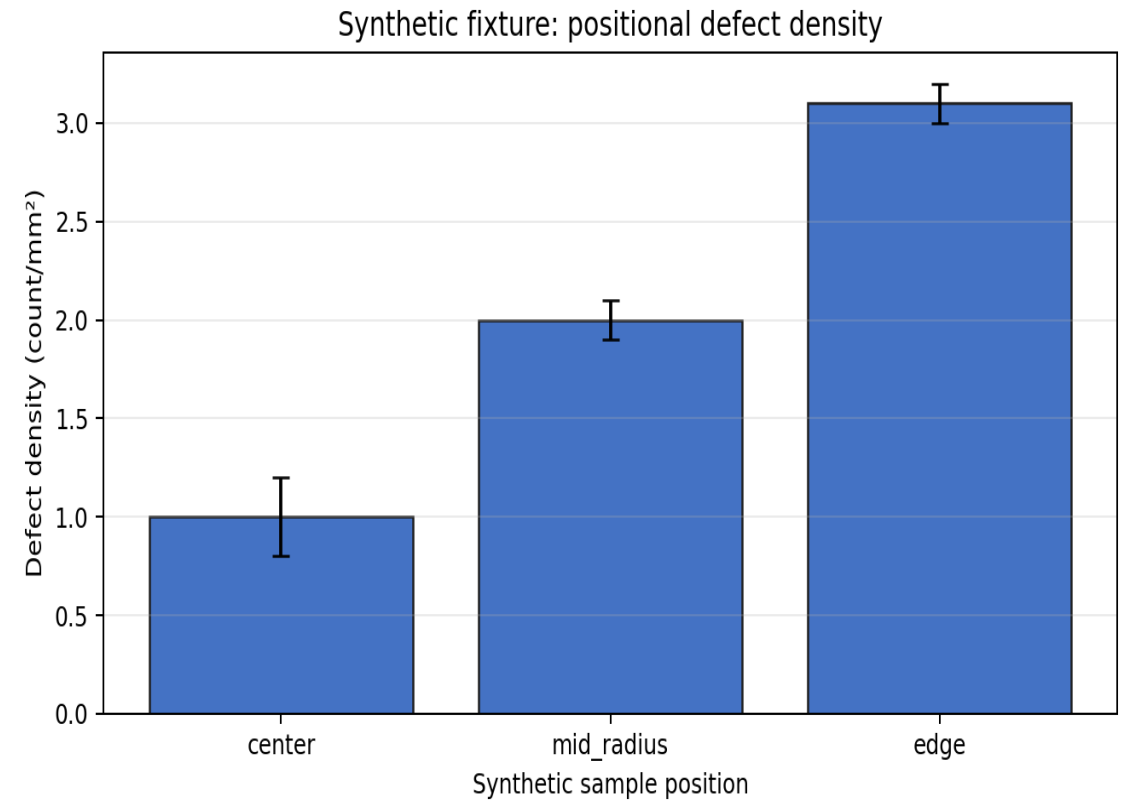
H001 | Results

RES101 | 平均導電度增加 $24\% \pm 5\%$ SD



H001 | Results

RES102 | 訊號 CV 僅下降 $4\% \pm 6\%$ SD , 屬 No-Go



H001 | Integrated Discussion

Supporting | RES101

Contradicting | RES102

Cross-experiment pattern | 導電度提升成立，但重複性沒有同步改善。

Mechanism assessment | bulk conductivity 是部分機制，不足以解釋 interface-sensitive variation 。

Alternatives | 接觸電阻；局部壓力差；電極幾何

Remaining uncertainty | contact interface 是否為主導因素仍需匹配導電度後檢驗。

H001 | Layer Summary / Decision

Answered | bulk conductivity 影響缺陷梯度，但不足以單獨解釋重複性。

Hypothesis status |
partial_support

Decision | Partial-Go: Bulk conductivity is contributory but insufficient.

Unresolved | interface contact 的貢獻尚未量化。

Next question | 匹配 bulk conductivity 後，contact resistance 是否主導變異？

Next Step | NS101

H01 → H02 | Hypothesis Transition

Previous hypothesis | C101

Key results | RES101, RES102

Not explained | 導電度增加後訊號重複性沒有改善。

New observation | E201

Therefore | 剩餘變異隨電極接觸位置改變，因此核心解釋由 bulk transport 轉向 interface contact。

New hypothesis | C201

H002 | Hypothesis

Hypothesis | Contact resistance drives instability at low pressure.

Falsifier | Matched conditions do not change the predicted CV or resistance.

Research question | 低接觸壓力下，contact resistance 是否主導訊號不穩定？

H002 | Problem

Problem | 導電度改善後重複性仍未改善，剩餘變異與電極接觸位置相關。

Previous finding | H01 支持導電度梯度存在；H01 未支持導電度單獨主導重複性

Conflict | bulk transport 與 interface contact 的相對貢獻尚未區分。

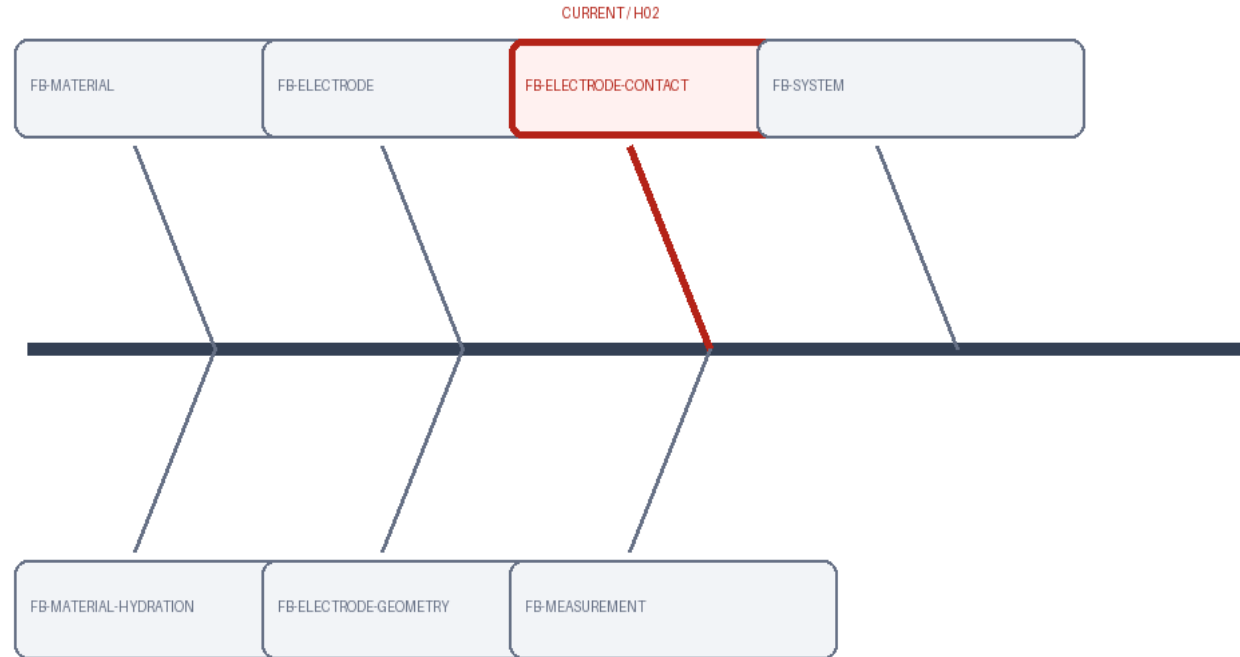
Research question | 低接觸壓力下，contact resistance 是否主導訊號不穩定？

Scope | 匹配 bulk conductivity，改變 contact pressure 的對照實驗。

H002 | Total Fishbone / Research

Map

Historical snapshot |
FB001 rev2
Focus branch | FB-
ELECTRODE-CONTACT
Immutable map; current
branch highlighted.



H002 | Observation → Literature → Mechanism

Observation | 位置依賴缺陷與訊號變異已觀察到。

Problem | 現有模型無法解釋此變異。

Research question | 低接觸壓力下，contact resistance 是否主導訊號不穩定？

Literature consensus | transport gradient 可產生位置效應。

Alternatives | interface contact remains alternative

Gap | 缺少控制比較隔離機制。

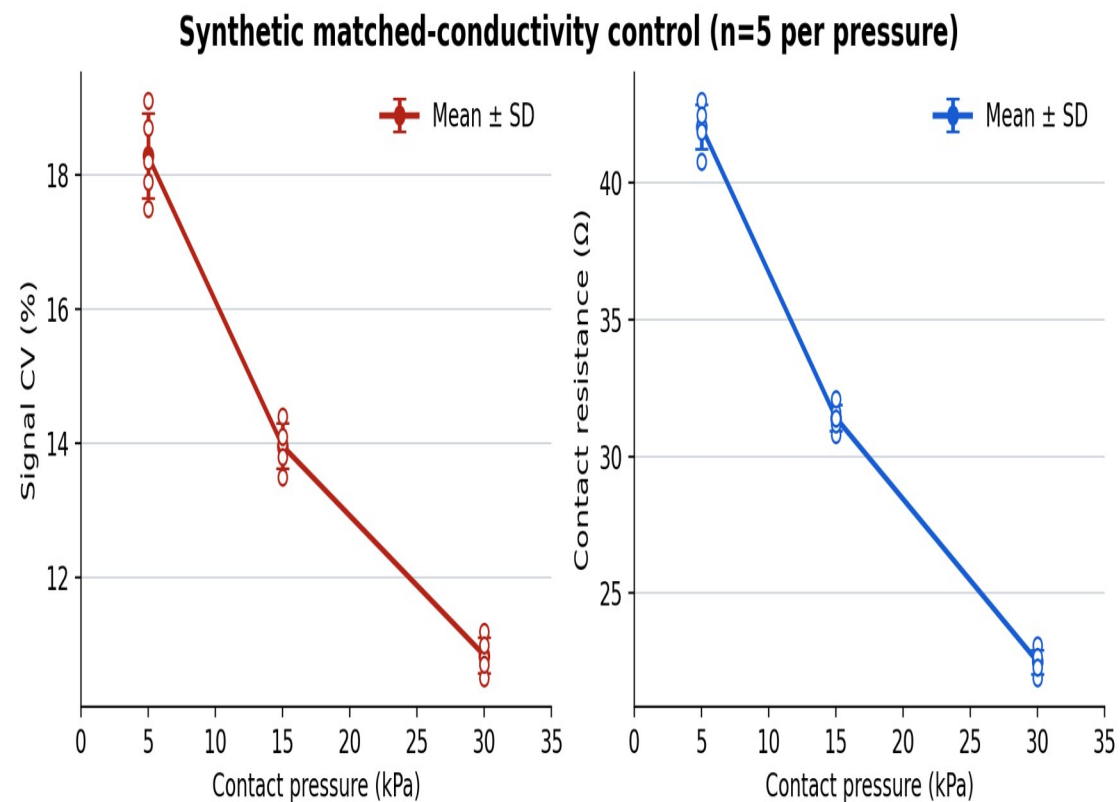
Implication | 需匹配條件後比較。

Mechanism | contact resistance

Strategy | 匹配導電度並改變接觸壓力

H002 | Results

EXP201 |
RES201 | 高壓條件 CV 下降 $38\% \pm 7\% \text{ SD}$,
contact resistance 同步下降



H002 | Layer Summary / Decision

Answered | 低接觸壓力下
contact resistance 是主導不穩定
來源。

Hypothesis status | supported
Decision | Go: Contact-pressure
control discriminates the interface
mechanism.

Unresolved | 高壓循環的耐久性
與 wear 尚未測試。

Next question | 接觸壓力改善能
否在長期循環下維持？

Next Step | NS201